Comparative Study of the Working of Various AI Classifiers for identifying Ragas of Indian Classical Music

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Abstract- **Music is an art of melody and sound. Classical music provides a framework for creating melodies and musical compositions. The most popular methods are Western classical music and Indian Classical Music (ICM). In ICM, the melodic pattern is called a Raga. Each Raga is classified based on the notes present, the order of the notes and the emotion (Rasa). Identifying a Raga is necessary for various music analyses. The objective of this research paper is to conduct a Systematic Literature Review (SLR) specifically to compare the working and performance of all the classifiers used in AI models proposed in previous research. The work involves a systematic literature review and a meta-analysis of all the AI classifier models used in ICM Raga Identification. The purpose of this study is to provide knowledge of the best performing AI classifier in Raga identification of ICM. The detailed analysis indicates that the model using Time Delay Neural Network (TDNN) displays the best performance over a large data set. This classifier can be considered for all further studies involving Raga identification which still in its nascent stage of research.**

Keywords- Raga, Rasa, Notes, Indian Classical Music, Systematic Literature Review, Time Delay Neural Network

I. Introduction

Music is a fluid and creative field where only melody and emotions should matter. However there are standard methods which help us learn and generate these melodies. These methods have been taught and learnt since thousands of years as a quest to generate great melodies as mention in the review work [22]. Such a framework is called as Classical Music. There are many styles of classical music worldwide. Until very recently amalgamation of music and computation was not common place. However the advancement of technology leads to the term computational musicology [8], where computing algorithms are used for various aspects of music. With the advent of AI, in the recent years AI is being used in many fields of music from composing to analyzing musical clips. From year 2000 and onwards we see a steady boost in the research work involving concepts of AI in music, as seen in one the analytical comparison work of Bhat et.al. in research work [3] and Biswa et.al.in their study [5] comparing deep learning methods. Usage of AI in Indian Classical Music is increasing and currently there is a lot of scope to use AI tools for various aspects of music.

One of the aspects of music is classification of music, in terms of the tune (Raga) or emotion (Rasa). In the recent years i.e. from 2021 till date the research works of music classification using AI has gone up by almost 164.7%, to mention a few [1],[3],[5],[8],[45] are works during this time. Hence there is a need for in-depth study of these research works. Such a study will be hugely helpful to form the base of any future classification works. It is seen that any Raga identification technique mainly involves feature extraction from a given music audio clip and classification either using an algorithm or an AI classifier. Many works and even comparative studies of these works have happened in this field. However the number of Ragas present in any stream of ICM is humungous The comparative studies such as [12],[4],[3],[1] are generic and limited to one or 2 methods only. Also the number of unique Ragas has not been considered as a key point of the analysis. Hence this study is an endeavour to focus on these shortcomings. This analysis can be of help for further research in identification and classification of classical music audio.

II. Related Work

There have been similar reviews in the field of music analysis using AI models, many recognition works like [34],[17] are real time systems without AI and few comparative works of other related music analysis as seen in [20],[35],[36], [37], [39] uses ML techniques for emotion classification. Research work [48] is a ML technique to detect emotions by facial expressions, while [49] is a work ML technique for identification of Deep Fake. [40] is a review of techniques in music selection in healthcare, another closely related review article involving music analysis. Most of the reviews and surveys emphasise on the latest and the best techniques based on various performance factors. Lazardo Moyisis et. al. in his work [12] has conducted a review on Deep Learning methods used in Music Signal Processing. It explains methods like CNN, RNN, LSTM and GAN in the area of MIR Systems. Biswa et. al. in the research [4] is a comprehensive comparison of CNN, RNN and XGBoost

the research concludes that RNN shows the best results. Satwik Acharya et. al. in [3] explore the techniques like ANN, LSTM and XGBoost models in [1]. In [3] Bhat et. al. conduct an analytical comparison of 4 methods ANN, CNN, Bi-LSTM and XGBoost for instrumental audio. It may be noted that all the

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studies focus on the overall performance of the model and the unique Ragas in each dataset has not been considered in the evaluation criteria. A model which displays 100% performance result with only few distinct Raga may fail when more number of ragas especially sounding similar are included in the dataset. This study specifically analyses the performances of the classifiers used in the various AI models in music classification for ICM. It focusses on two factors, the number of unique Ragas and the performance. This study uses SLR method and meta-analysis for analysing the performance of the classifiers.

A detailed literature survey was conducted in the pursuit of understanding all the past work in identification and classification of Ragas using AI. The details of the previous work are as shown in chart Fig. 1.

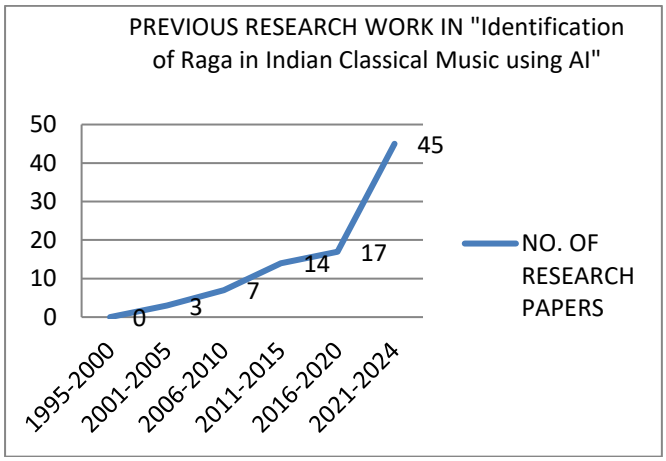

Fig. 1. Systematic Literature Review

It is evident from the chart in Fig.1 that no research work involving Indian Classical Music Raga identification using AI has been published till the year 2000. The number of papers steadily increases. A detailed study of all the models and classifiers from year 2021 and beyond is considered for this paper, the study uses SLR method and a meta-analysis is presented. Similar comparative study has been conducted on analytical comparison of classical models for raga identification. There are no studies which involve meta-analysis; there is a need of in-depth meta-review of past work to bridge this gap. This proposed research paper is an endeavor to overcome this research gap.

III. Methodology

The proposed study is a detailed Systematic Literature Review and Meta-Analysis of AI Classifiers with relevance to Classical Music Identification. The SLR has been conducted based on the knowledge acquired from research works on scientific SLR. The steps of SLR have been obtained by a similar work for scientific literature review i.e. [11]. Inputs for the SLR methodology is also followed from [16]

A. Research Question

What according to the previous research methods is the best AI classifier for Raga/notes identification in Indian Classical Music? How many unique Ragas are identified by the AI classifiers in each of these methods? What is the performance of each of these methods?

B. Steps for the analyses:

An in-depth study and analyses based on 2 factors have has been conducted and a final interpretation of the best classifier method most suitable for Classification of Ragas in ICM is reported.

Fig. 2. Steps for the Meta Analysis

The methodology used for the analysis is mentioned in Fig.2.

C. SLR Methodology:

Keywords "Artificial Intelligence + Indian Classical Music" were the main key words used. The search was systematically narrowed down as shown in Fig. 3.

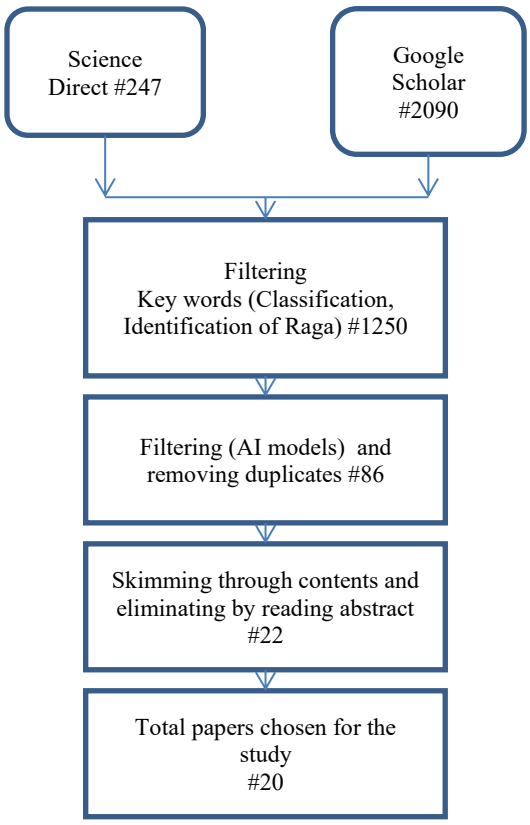

Fig. 3. SLR steps for choice of Literature

The chosen papers are review articles and papers introducing new techniques.

D. The Generic AI Model

This study focusses on the classifier module mentioned in Fig.4. [22]Most models have used one of the AI classifiers for Classification namely, CNN, TDNN, SVM, RNN or Hybrid classifiers. Although the overall performance of a model

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depends on both feature extraction technique and classifiers, this study is confined to only the performances of the classifiers mentioned in the various research models.

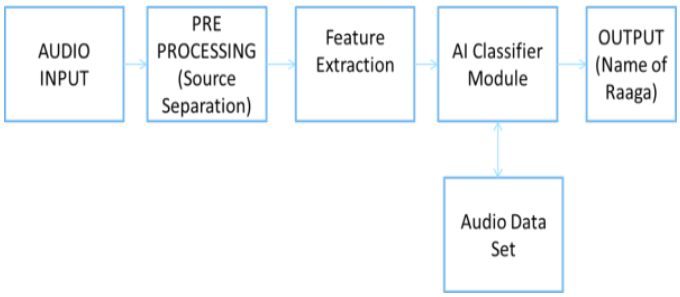

Fig. 4. Generic model for AI based Raga Classification

IV. Pre Existing Classifiers and Analysis

An in-depth study of the pre-existing classifiers in various relevant classifications was conducted. This section describes preexisting classifiers, key factors considered for the analysis of classifiers and the detailed analysis of the classifiers considered in preexisting work for ICM Raga identification. The analysis followed by an interpretation concludes the study.

A. *Pre Existing classifiers*

It is seen that CNN, RNN, SVM, TDNN and hybrid classifiers are used in many varied research works involving various types of classifications. This section mentions a few of them.

1) *CNN*

Many studies have used the concept of CNN for various classifications. CNN architecture has multiple convolutional layers followed by a max pooling layer and a normalization layer. To mention a few, works of [[32]] is a predictive analysis of agricultural waste, [[33]] involves sentiment analysis, [[38]] is for classification of musical instruments, [46] is meant for speaker identification of an audio, research work [43] is work on image restoration. Convolutional Neural Networks is very efficient method used for image classification which can be used in different types of applications.

2) *TDNN*

A Time Delay Neural Network is a 1D Convolutional Neural Network. They are specifically designed to deal with time varying inputs which make it ideal for speech recognition and other sound related applications.[50]. using TDNN method for language identification is one such example.

3) *RNN*

Recurrent Neural are suitable for handling temporal sequences. RNN uses an internal memory state and can process variable length inputs. Speech recognition software often uses RNN models, research work [47] is one such study which uses Bi-LSTM RNN for spoken word recognition. The research work mentioned in [42] uses RNN for earthquake detection.[30] uses RNN for a music genre recommender system.

4) *SVM*

A Support Vector Machine is a very widely used algorithm for both linear and non-linear classification. They are used for both binary and multi-class classification. SVM is a preferred classifier in many AI classification models. One of the classifiers mentioned in [27] is an SVM classifier used for labelling moods in different Hindi song clippings. The dataset consists of 1000 different songs of different moods.

5) *Hybrid*

These models are multi-layered and use one of the above mentioned classifiers in each layer. Model mentioned in [41] is a hybrid model using 7 different layers for pose recognition in yoga. Similarly [44] is a hybrid CNN-BiLSTM model used for recognition of speech segment end points in speech processing, [23] is uses CNN-BiLSTM model for music emotion classification.[28] is a research work using a combination of SVM and Bayesian classifiers to recognize the musical instruments in a ICM audio clip. A consolidation of the preexisting works is mentioned in Table I.

TABLE I. PREEXISTING WORKS AND IT'S CORRESPONDING CLASSIFIERS

Sl. No	AI Classifier	Research Work
1	CNN	Most common approach, the audio is converted into signals and the classifier uses image processing techniques for identification.
2	TDNN	It is a special 1D-CNN,most suitable for the analyses of sequential data along the time axis, extensively used in speech recognition
3	Support Vector Machine	used in both binary and multi class classification. It can handle both linear and non-linear data.
4	Recurrent Neural Network	Most suitable in handling temporal sequences.
5	Hybrid	Multi layered classifiers where each layer has a different classifiers like CNN, RNN or SVM

B. *Key points considered for the analysis of classifiers for Raga identification in ICM*

Based on the study of the pre-existing work, three key points are considered for the analysis. The key points considered for the analysis of the classifier are based on the performance and the number of unique Ragas present in the dataset. Each classifier is analyzed and evaluated with respect to the key points mentioned in Table II.

TABLE II. KEY POINTS CONSIDERED FOR ANALYSIS OF CLASSIFIERS FOR RAGA IDENTIFICATION IN ICM

Sl. No	Key Point	Analysis
1	Type of ICM	HCM/CCM
2	No. of Ragas in the data set	Number of distinct Ragas
3	Performance Accuracy	Accuracy of identification in percentage

C. *Detailed Analysis of the classifiers for ICM Raga identification.*

This section is confined to the discussion of a detailed analysis of the classifiers used in ICM Raga identification emphasizing on the key factors considered for analysis.

1) *CNN*

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[20]is a CNN-LSTM model. It takes a chromagram as an input, this is rendered to a multi-layered CNN, a unique reshape layer is follows and a feature vector of reduced dimension passes through a LSTM layer and finally the output is decided by the fully connected max pool layer, the unique Ragas in the dataset is 12 Ragas of HCM. Research work [[18]] uses CNN based sliding window principle. The input for this model is a mel-spectogram. The uniqueness of this model is the fact that it does not use pitch extraction or metadata for identification, it works on a data set of 15 Ragas of CCM. Research work [2]uses a CNN model with Swaragram[[25]] as an input, it has multiple convolutional and batch normalization layers and a max pooling layer, it uses feed forward layer that handles meta data. Finally a fully connected layer followed by a Softmax layer presents the final identification, the dataset contains only 8 unique Ragas of HCM. Study [[13]] is multimodal using both video and audio inputs for identification purpose. It uses separate K Kernel convolutional layers for audio and video analysis, audio softmax and video softmax layers are finally combined by Fusion Prediction layer which decides the name of the Raga, limited to only 2 Ragas of HCM.[7] is a multimodal classification from pose and audio time series extracted from, the model uses an inception block and separate audio and video max pooling layers using a dataset of 9 different Ragas of HCM.[26] uses a specialized Feed-Forward neural network for tonic identification on a dataset of 17 Ragas of HCM.

2) TDNN

Research work [14] is the study which has performed very well on a very large dataset as compared to the others. The feature extraction uses a unique method of creating custom bins. A TDNN can identify temporal dependencies in sequential data. The model contains N-layer TDNN followed by M-layer LSTM leading to a Softmax layer which identifies the Raga, the data set has 172 unique Ragas of CCM.

3) SVM:

Research work [10] uses SVM with RFE (Recursive Feature Elimination) for improving the performance. The input to this classifier are time and frequency domain features, a multi kernel SVM is used for classification, followed by an RFE method for further enhancement of the classification. This method uses a dataset of 8 Ragas of HCM. Study [6] uses SVM parameter optimized model for rasa identification, the model gives good results in identifying the 9 Rasa with the input audio of 816 different tunes. However it does not identify a Raga in particular

4) RNN

[[9]] proposes a Recurrent Neural Network Model which essentially has a series of interconnected layers where the output of one layer is passed on to the input of another layer. LSTM memory networks are used for refining the accuracy and validation. This study has only 2 unique Ragas of CCM in the dataset. Research work [[8]] uses Bi-directional LSTM, the model has a self-feedback hidden layer and uses memory cells. The memory cell has 3 gates namely the input gate, forget gate and the output gate. This model uses hyperparameters for optimization. This model has used 9 Ragas of CCM and 7 Ragas of HCM in it's dataset.[31] uses regression for classification of 72 Melakarta Ragas of CCM.

5) Hybrid Models

[[15] uses stacked ensemble mode which uses multi layered classifiers, research work [[19]] is a unique comparative study of three models multi-layer CNN, multi-Layer RNN and SVM. Research work [24] uses a hybrid model of LSTM +SVM for music genre classification on a dataset of over 1000 tunes. [29] is a work of classifying 15 different Ragas of CCM using RNN+LSTM models and a hybrid ANN model.

V. Interpretation and Results

The consolidation of all the classifiers and their performance and number of unique Ragas in the dataset has been depicted in Table III. The analysis is as follows. From the analysis in Table. III., we can imply that CNN is a classifier used in many models, although it achieves a best performance accuracy of 98.22%, the number of Ragas in the dataset is very minimal. Performances of 98.22% is noticed in model of [13] using CNN classifier but however it is limited to 2 Ragas only. [9] uses an RNN classifier but limited to 2 distinct Ragas only. [31] uses RNN classifier and achieves a performance of 100% for 72 Ragas. [15],[19] use hybrid models to achieve 97% and 98.3% respectively over a dataset of 12 and 64 Ragas respectively. [14] uses TDNN classifier with a performance of 95.3% on a varied dataset of 172 unique ragas.

TABLE III. Consolidation of the analysis based on the key factors

Sl. No	AI Classifier	Research Work	Key point #1 (Type of ICM)	Key Point #2 (No. of unique Ragas in Dataset)	Key point #3 (Performance Accuracy)
1	CNN	[[20]]	HCM	12	83%
		[[1]]	HCM	8	89%
		[[18]]	CCM	15	72%
		[[13]]	HCM	2	98.22%
		[[6]]	HCM	9	85%
		[[26]]	HCM	17	95%
2	TDNN	[[14]]	CCM	172	95.31%
3	SVM	[[5]]	Generic	816	77%
		[[10]]	HCM	8	80%
		[[27]]	Generic	1540	76%
4	RNN	[[9]]	CCM	2	96%
		[[8]]	HCM + CCM	7 + 9	94%
		[[31]]	CCM	72	100%
5	Hybrid	[[15]]	CCM	12	97%
		[[19]]	CCM + HCM	28 + 36	98.3%
		[[24]]	Genric	1000	89%
		[[29]]	CCM	15	86%

From the above analysis we can interpret that TDNN method is able to show an impressive 95.3% on a varied dataset of Carnatic Classical Music with 172 unique Ragas. Although the other classifiers have shown better results, the dataset is limited to a small number of unique Ragas in the dataset. The results of other models can be observed in Table III. It may be noted that TDNN is a special 1D-CNN which displays the best performance.

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VI. Conclusion and Future Scope

This SLR study intended to focus on all the work involving ICM Raga Identification with AI models and concluding the model which exhibits the best performance over a large dataset. The findings of this study are that a TDNN classifier, which is a 1D-CNN, has shown the best performance with the maximum number of unique Ragas in the dataset.

Music and AI are a very vast area of research. Not many tools and applications are available for identification of Raga dynamically. Studies of this scope would largely help the researchers and also the world of music. The intent of this study is to provide an inference to pick up the work which has shown best results on large unique dataset. On the basis of this study, TDNN classifier will be an ideal choice for further improvisation and any new applications in the field of Raaga identification, mood analysis or MIR.

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